Notes 2

1. What is an Operating System?

An operating system provides all fundamental software features of a computer. An OS enables you to use the computer's hardware providing you the basic tools that make the computer useful. All of those features relay on the OS's kernel. Other OS features are owed to additional programs that run atop the kernel.

2. What is a kernel?

An OS kernel is a software component that's responsible for managing low-level features of the computer, including the following managing system hardware, memory allocation, CPU time, and program to program interaction.

3. Which other parts aside from the kernel identify an OS?

Aside from the kernel, an OS is identified by things like system libraries, APIs, and the user interface (command line or GUI). These let programs and users interact with the system. Also, each OS has its own internal design for handling files, memory, and devices, which makes it unique.

4. What is linux and linux distribution?

Linux is a Unix-like operating system made up of a kernel, libraries, and utilities. It is open-source, free to use, and very popular in both academic and business environments. Linux is highly scalable, customizable, and runs on almost any system—from phones to supercomputers. Many servers and applications on the internet also run on Linux.

A Linux distribution (distro) is a complete package of the Linux system. It includes the Linux kernel plus core Unix tools, supplemental software, startup scripts, and an installer. Different distributions (like Ubuntu, Fedora, Red Hat, Debian, Arch, etc.) may use different versions of the kernel, tools, or desktop environments, but all are based on Linux.

5. List at least 4 linux characteristics:

1. Open Source Software – anyone can view, edit, and share the code.
2. Free of Charge – most Linux systems are available without cost.
3. Highly Scalable & Customizable – can run on small devices up to supercomputers.
4. Secure & Reliable – widely used for servers and critical systems.

6. What is Debian?

Debian is one of the oldest and most stable Linux distributions. It is community-developed and serves as the base for many other distros, including Ubuntu. Debian focuses on stability, free software principles, and a wide selection of software packages.

7. List and define the different types of licensing agreements

- Proprietary License – software owned by a company; users can use but not change or share (e.g., Windows).
- Free Software License (GNU/GPL) – allows users to run, study, share, and modify the program.
- Open Source License – source code is available, but terms may vary (MIT, Apache, BSD).
- Shareware – software is free for trial but requires payment for full features.
- Public Domain – no copyright, completely free for anyone to use.

8. What is Free Software? Define the 4 freedoms.

Free software means users have the freedom to control the program, not just use it. The 4 freedoms are:

1. Freedom 0 – Run the program for any purpose.
2. Freedom 1 – Study how the program works and change it (requires access to source code).
3. Freedom 2 – Share copies with others.
4. Freedom 3 – Improve the program and release improvements to the public.

9. What is virtualization?

Virtualization is the process of creating a virtual version of a computer system (like hardware, operating system, storage, or network) so multiple operating systems or applications can run on one physical machine.